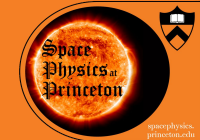


GEMINI Test Plan 4/23

William Li - Junior Independent Work - Spring 2026
Space Physics at Princeton

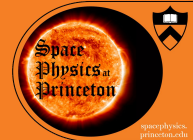
Proposed Tests



3 sets of data will be gathered for each beam: (1) Baseline, (2) $0.6 \mu\text{g}/\text{cm}^3$ carbon foil, (3) graphene foil

Species	Energies	Deflector Plate Target Voltage	Deflector Plate Voltage Range	Priority	Purpose
p+	500 eV 750 eV 1 keV 1.25 keV 1.5 keV	$\pm 542 \text{ V}$ $\pm 813 \text{ V}$ $\pm 1084 \text{ V}$ $\pm 1355 \text{ V}$ $\pm 1626 \text{ V}$	456-629 V 684-943 V 912-1257 V 1140-1570 V 1367-1884 V	Highest	Measure angular scattering below 1 keV as a function of foil type and beam energy
p+	500 eV 750 eV 1 keV 1.25 keV 1.5 keV	+5 V (North) 0 V (South)	Manually tune via small blue power supply (min resolution 0.1V desired)	High	Measure secondary electron emission below 1 keV as a function of foil type and beam energy

Angular Scattering Deflector Plate Voltage Limits



Absolute Maximum Limit to set in software: ± 3.8 kV

- High Voltage Qualification Testing held -3.8 kV from 11:47 AM on 4/17 to 10:26 AM on 4/18
- Since only testing beam energies of 1.5 keV or lower, the maximum anticipated voltage is 1884 V (see Slide 2)

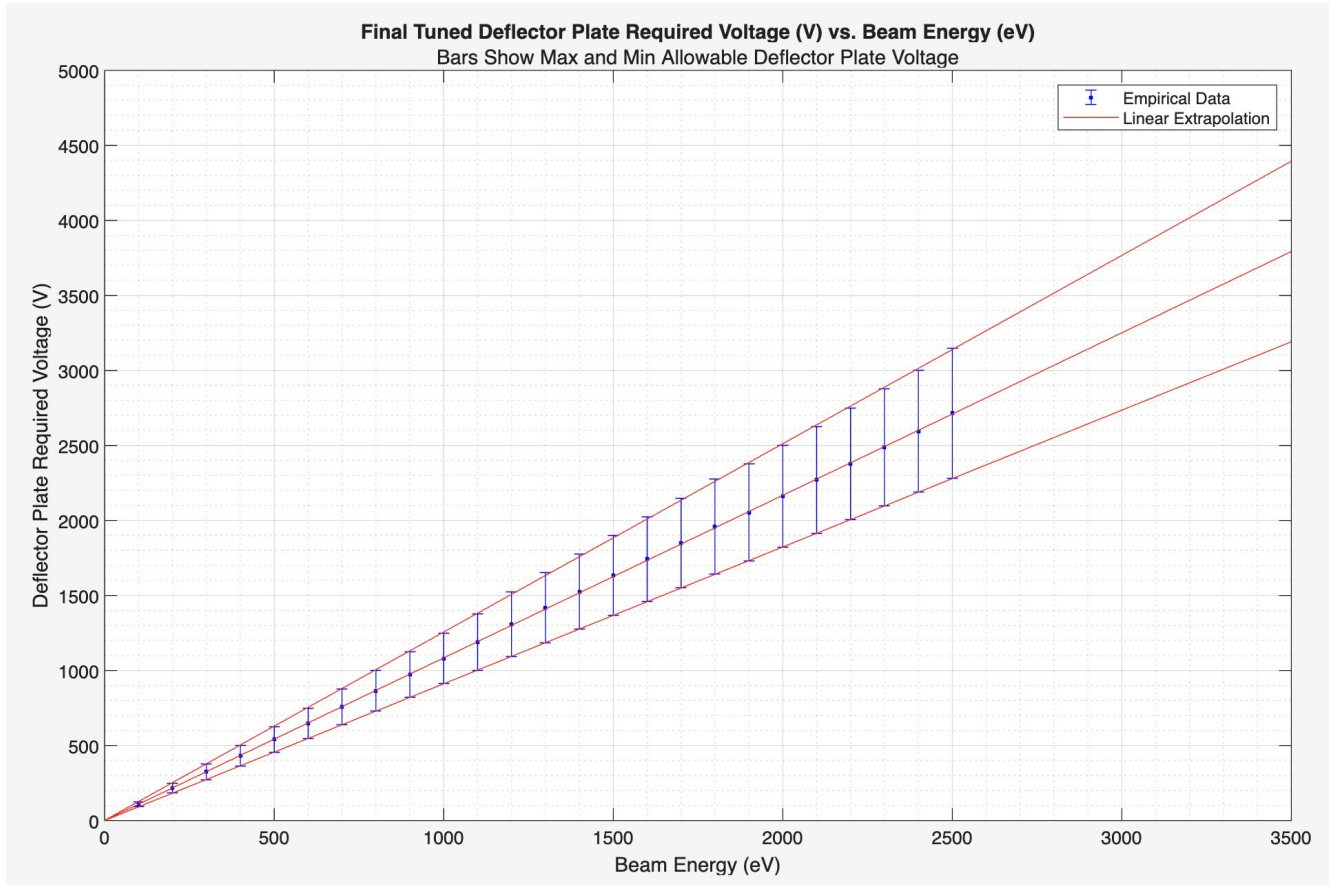
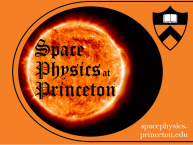
Other High Voltage Qualification Test Results

- 4/8 16:38 - 17:34 at -4.5 kV (56 minutes before trip)
- 4/8 19:07 - 19:29 at -4.5 kV (22 minutes before trip)
- 4/9 09:06 - 13:36 at -4.5 kV (4 hours 30 minutes before trip)
- 4/9 13:56 - 14:16 at -4.5 kV (20 minutes before trip)
- 4/9 16:30 - 4/10 02:11 at -4 kV (9 hours 41 minutes before trip)

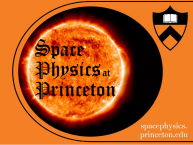
Reworked MCP Bracket and polished Deflector Plates

- 4/17 17:43 - 19:28 at -4 kV (1 hr 45 minutes before trip)
- **4/18 11:47 - 4/19 10:26 at -3.8 kV (22 hrs 39 minutes before trip)**

Angular Scattering Deflector Plate Voltage Lookup Graph



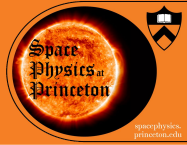
Angular Scattering Deflector Plate Voltage Lookup Table



Energy	Target Voltage (V)	Min Voltage (V)	Max Voltage (V)
500 eV	± 542	456	629
750 eV	± 813	684	943
1 keV	± 1084	912	1257
1.25 keV	± 1355	1140	1570
1.5 keV	± 1626	1367	1884

Deflector Plate voltage is only a function of beam energy, not species

Dwell Time



Concern: Pinhole will result in high counts/area on the face of the Quantar MCP

Pinhole area was not measured prior to installation in the Science Chamber but has a minimum diameter of 1.75mm based on drill size used during manufacturing

Pinhole Diameter: 1.75 mm $\rightarrow$ 2.4 mm² Pinhole Area

MCP Diameter: 40 mm $\rightarrow$ 1256.6 mm² MCP Area

Count rate on MCP = (MCP Area/100)*Pinhole Area = 30.2 Hz

- 100 based on MCP allowable parameters?

If t = dwell time, precision = $\sqrt{N}/N = \sqrt{30.2\text{Hz} \cdot t}/(30.2\text{Hz} \cdot t)$

- $t = 120\text{s}$ gives 1.7%